

# DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery

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## Abstract

Analyzing street-view imagery with computer vision models for rapid, hyperlocal damage assessment is becoming popular and valuable in disaster resilience research and management, but existing approaches typically rely on “black box” pre-trained vision models, which lack interpretability and reliability. This study proposes DamageArbiter, a multimodal disagreement-driven arbitration framework powered by Contrastive Language-Image Pre-training (CLIP) models to improve the accuracy and robustness of damage assessment from street-view imagery. **DamageArbiter leverages the complementary strengths of unimodal and multimodal models, employing a lightweight logistic regression meta-classifier to arbitrate disagreements and applying post-hoc confidence calibration through temperature scaling to correct overconfidence.** Using 2,556 post-disaster street-view images, paired with manually generated or large language model (LLM)-generated text descriptions, we systematically compared the performance of DamageArbiter with unimodal models (image-only and text-only) and CLIP-based multimodal models. Results show that DamageArbiter improved accuracy to 75.85% and the Matthews correlation coefficient (MCC) to 0.619. These results surpass the text-only baseline (accuracy 63.07%, MCC 0.413) and the image-only baseline (accuracy 74.33%, MCC 0.595). **Meanwhile, image-only models made many overconfident errors (70.58%). Post-hoc confidence calibration reduced these errors to 51.37% without lowering accuracy, showing that accuracy alone is not enough, and overconfidence should be measured and corrected.** Overall, this work advances street-view-based disaster assessment from coarse severity classification toward a more reliable framework.

Keywords: Contrastive Language-Image Pre-training (CLIP), Street-View Imagery, Vision-Language Models, Multimodal Learning, Damage Assessment

## 1. Introduction

Timely and accurate post-disaster damage assessments play a pivotal role in supporting emergency response, guiding resource allocation, and informing short-term and long-term recovery efforts. Among the

diverse data sources employed in disaster damage assessment, street-view imagery has emerged as a valuable source for high-resolution, human-perspective observations of structural and infrastructural damages that are often difficult to capture from aerial or satellite imagery (Lozano & Tien, 2023; Yamazaki & Matsuoka, 2007; Ye et al., 2024). For example, post-disaster street-view images can reveal fine-scale damage indicators, such as fallen utility poles, building watermarks, inundated streets, debris, damaged roadways, and fallen trees. These visual signals provide critical evidence for assessing localized impacts of disasters. Consequently, there is growing interest in both research and practice in leveraging street-view imagery for post-disaster damage assessment.

Traditional approaches to street-view-based damage assessments typically use manually labeled datasets to fine-tune pre-trained computer vision models, such as the Visual Geometry Group network (VGG), the Shifted Window Transformer (Swin Transformer), and ConvNeXt, to classify disaster impacts into different levels (H. Li et al., 2024; Xue et al., 2024; Zhai & Peng, 2020). However, several critical limitations exist. First, these methods rely on extensive manual annotations, which are costly and time-consuming, making them slow to adapt to new or evolving disaster scenarios. Second, even when large, annotated datasets are available, the labels are often limited to coarse categories, typically three or four damage levels, providing limited insights into the underlying causes and responsive actions. For example, an image labeled as “severe” may reflect the destruction of a large building, the collapse of critical infrastructure (e.g., power poles), or both. Third, reasoning results from these models remains difficult. Previous work has leveraged methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) to explain how vision models interpret disaster damages from images (Selvaraju et al., 2017; Yang et al., 2025). Grad-CAM identifies spatial regions within an image that contribute most to a model’s prediction, but it lacks the capacity to convey fine-grained semantic information such as the specific damage type, severity, and its cascading effects (e.g., a power outage may cause school closure), which are essential for actionable decision-making.

Recent advances in vision-language models (VLMs) open promising avenues for disaster damage assessment using street-view imagery. By aligning images with descriptions, VLMs can capture richer semantics from images, supporting fine-grained reasoning and generating interpretable outputs. In particular, the Contrastive Language-Image Pretraining (CLIP) model provides a general framework for cross-modal alignment that can be adapted for disaster scenarios. For instance, Yang and Zou (2025) demonstrated the potential of VLMs in perceiving multidimensional disaster impacts from street-view images. However, the application of VLMs in disaster damage assessment remains in its early stages, primarily due to the lack of high-quality disaster-related image-text pair datasets for fine-tuning and benchmarking VLM models (Jayati et al., 2024; Wang et al., 2024). Further research is needed to examine whether VLMs can enhance street-view-based damage assessment and to systematically evaluate their reliability, interpretability, and decision-support utility.

This research proposes DamageArbiter, a disagreement-driven, CLIP-enhanced multimodal arbitration framework for damage assessment from street-view imagery. A key novelty is the disagreement-driven arbitration mechanism, which dynamically resolves conflicts between visual and textual evidence to address ambiguity in multimodal damage assessments. The overarching objective is to develop the DamageArbiter framework and benchmark its accuracy, uncertainty, and explainability against baseline unimodal models, including text-only or image-only approaches, as well as multimodal models. In this work, uncertainty is evaluated through overconfidence, defined as the systematic tendency for a model to assign high confidence

to its predictions even when those predictions are incorrect or uncertain. Post-disaster street-view images collected after the 2024 Hurricane Milton, along with textual descriptions generated by human annotators and large language models (LLMs), were used to fine-tune the models and benchmark their performance. LLM-generated descriptions are compared with human annotations to evaluate when automated descriptions can support robust disaster assessment. We validated the models' effectiveness and deployed the optimal configuration. This study advances street-view-based disaster assessment by shifting from black-box prediction toward a more transparent, interpretable, and trust-aware framework suitable for high-stakes disaster response.

## **2. Related Work**

### **2.1 Analysis of Street-View Imagery for Disaster Damage Assessment**

Street-view imagery has become an increasingly valuable data source for fine-grained, human-perspective urban analytics. These images can capture street-level details, such as building characteristics, the availability and quality of green space, and infrastructure conditions. Platforms such as Mapillary, Google Street View, and OpenStreetMap (OSM) enable widespread access to such data at different locations and timestamps, facilitating large-scale, hyperlocal, and longitudinal urban analysis at unprecedented scales (Alvarez Leon & Quinn, 2019; Mahabir et al., 2020; Sánchez & Labib, 2024). Meanwhile, advances in deep learning-based computer vision have significantly accelerated research in visual-based investigations. Rather than training models from scratch, researchers have fine-tuned pre-trained visual backbones, such as VGG, Swin Transformer, and ConvNeXt, leading to notable gains in image-based classification accuracy and computational efficiency (Gupta, 2021; Ding et al., 2023; Xin et al., 2024). Swin Transformer leverages hierarchical attention mechanisms to capture both local and global visual patterns (Liu et al., 2021, 2022), while ConvNeXt employs large-kernel convolutions to enhance the representational capacity of images beyond traditional convolutional neural networks (CNNs) (Feng et al., 2022; Woo et al., 2023).

Recent studies have begun leveraging pre-trained vision models for disaster damage assessment tasks (Mahabir et al., 2020; Xue et al., 2024; Zhao, Lu et al., 2024). For example, Gao et al. (2024) employed a YOLOv5-based computer vision framework to estimate First-Floor Elevation (FFE) from street-view images, providing a scalable alternative to field surveys and supporting flood mitigation planning in coastal regions. Wang et al. (2024) adopted a Mask Region-based CNN (Mask R-CNN) approach to extract visual features related to building structure, condition, and surrounding infrastructure, and used these indicators as proxies to estimate household-level vulnerability to natural hazards. Xue et al. (2024) fused street-view imagery with structured building information to assess post-hurricane building damage. H. Li et al. (2025) proposed CVDIsaster, a cross-view framework that combines street-view and very-high-resolution satellite imagery for mapping hurricane damage. Their framework achieved approximately 80% accuracy in geolocating disaster-related street-view images and 75% accuracy in assessing damage from those images. Yang et al. (2025) further showed that bi-temporal street-view imagery can improve hyperlocal damage estimation, with a dual-channel Feature-Fusion ConvNeXt model increasing accuracy from 66.14% for the Swin Transformer baseline to 77.11%.

## 2.2 Analyzing Street-View Imagery with Vision-Language Models (VLMs)

VLMs are multimodal models that jointly learn from visual and textual data to support cross-modal understanding, reasoning, and downstream applications. Conventional VLMs usually align image and text encoders with paired image-text data, using contrastive objectives to pull matched pairs closer in a shared embedding space and push mismatched pairs apart (Radford et al., 2021). Newer generative VLMs often connect a vision encoder to an LLM and refine the combined model through instruction tuning or supervised multimodal conversations (Li et al., 2023; H. Liu et al., 2024; OpenAI et al., 2024). Commercial systems include GPT-4o and later versions, as well as Gemini, while open-source or research-oriented examples include CLIP, BLIP-2, and LLaVA (Gemini Team, 2024; Li et al., 2023; H. Liu et al., 2024; OpenAI et al., 2024; Radford et al., 2021). Because of VLMs' capacity in connecting visual evidence with language representations, they can support tasks such as image-text retrieval, captioning, visual question answering, classification, and semantic reasoning.

Among VLMs, CLIP has become especially influential. CLIP learns aligned visual and textual representations through contrastive pre-training (Radford et al., 2021). CLIP-inspired methods have been applied to open-domain image geolocation (Haas et al., 2023), urban image retrieval and reasoning (L. Li et al., 2024), and cross-view disaster mapping with street-view and satellite imagery (H. Li et al., 2025). Previous work also enriched images with LLM-generated texts and input them into CLIP for downstream tasks, including street-view-based disaster impact investigations. For example, Jayati et al. (2024) paired ground-level building images with their corresponding ChatGPT-generated captions using VLMs, including CLIP and ViLT (Vision-Language Transformers), to assess post-disaster building damages. They reported that the CLIP model reached a validation macro-F1 score of 0.61, outperforming image-only ViT-B/32 (0.57) and text-only models (0.52). Li et al. (2025) proposed CLIP-BCA-Gated (CLIP with Bi-Cross-Attention and Adaptive Gating) to classify multimodal disaster-related Twitter/X messages, achieving 91.77% accuracy and 91.74% F1 on CrisisMMD (<https://crisisnlp.qcri.org/crisismmd.html>), a Multimodal Crisis Dataset.

These CLIP-related studies demonstrate the value of image-text alignment for disaster and crisis analysis, but they also leave important gaps. Most report classification performance without systematically examining the causes of erroneous predictions. In addition, many studies use automatically generated captions as semantic input without validating them against human-written descriptions or testing whether generated descriptions help or confuse model decisions. Furthermore, the reliability of CLIP-based damage assessment remains underexplored. Modern neural networks can be mis-calibrated, assigning high confidence to incorrect predictions even when uncertainty is warranted (Guo et al., 2017). In this study, overconfidence errors are incorrect predictions made with a large confidence margin, in which the model assigns a high probability to the wrong class. Ambiguity errors are low-margin cases in which the probabilities for competing severity classes are similar or where visual and textual evidence conflict. These concepts capture distinct reliability risks: overconfidence reflects false certainty, while ambiguity reflects unstable evidence. Both gaps are central to image-based disaster damage assessment, where visual cues can be subtle, partially occluded, or semantically ambiguous.

The DamageArbiter model developed by this study is poised to address the aforementioned limitations and challenges. **By improving predictive performance and reliability, the study aims to advance street-view-based disaster assessment toward a more transparent and operationally useful workflow.**

### 3. Data and Study Area

The Milton street-view imagery dataset (Milton-SV) was used for the experiments. It was collected after Hurricane Milton in 2024 in Horseshoe Beach, Florida, United States, and was constructed by Yang et al. (2025). The dataset contains 2,556 post-disaster street-view images that were manually classified into three damage severity levels: mild (658 images), moderate (1,196 images), and severe (702 images). Figure 1 shows the locations and damage labels of each street-view image included in the Milton-SV dataset. The manual labeling process was conducted by multiple trained annotators using predefined criteria based on observable disaster impacts, including fallen trees, building debris, damaged infrastructure, and inundated roads. This human annotation closely mirrors the way humans perceive and interpret disaster scenes and aligns with CLIP’s core objective of learning image-text semantic alignment. A cross-validation process was subsequently performed to ensure annotation consistency and accuracy.

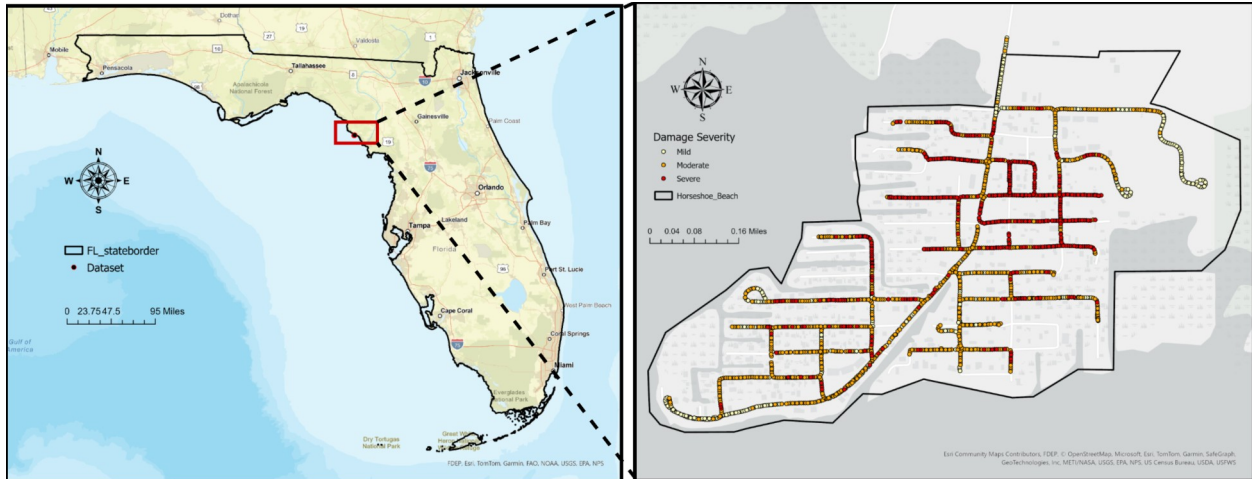


Figure 1. Hurricane Milton Damage Mapping and Street-View Sample Distribution

In parallel, we conducted a fine-grained descriptive annotation to enhance the interpretability and semantic depth of disaster assessment using automated annotation and human captioning. For automated annotation, we tested two cost-effective LLMs, GPT-4o-mini and Gemini 2.0 Flash-Exp, given their ability to balance annotation quality, inference speed, and computational cost for large-scale image labeling tasks. The automated annotation process was guided by a structured prompt designed to constrain the model’s behavior and reduce hallucinations. Specifically, the model was instructed to act as a disaster-damage annotator and describe only damage evidence explicitly visible in each image. The prompt emphasized factual, image-grounded descriptions and prohibited speculative or inferred content. The model was required to generate a concise damage description of approximately 50 words, focusing on observable damages, including fallen trees, damaged buildings, debris accumulation, and flooding conditions. All outputs were constrained to a predefined JSON schema to ensure structural consistency and facilitate downstream parsing. The generated captions from each LLM were compared with the original street-view images using CLIPScore (Radford et al., 2021), and the higher-scoring outputs were selected as the textual dataset for the subsequent experiments. GPT-4o-mini produced the higher-scoring outputs and was therefore used in the final LLM-caption setting for the experiments.

After descriptive annotation, each street-view image sample is paired with human- and LLM-generated text descriptions, enabling a systematic comparison of manual and automated perception of the same visual scene (Figure 2). This dataset design supports robust classification experiments, semantic interpretation, and evaluation of textual data’s role in disaster assessment.

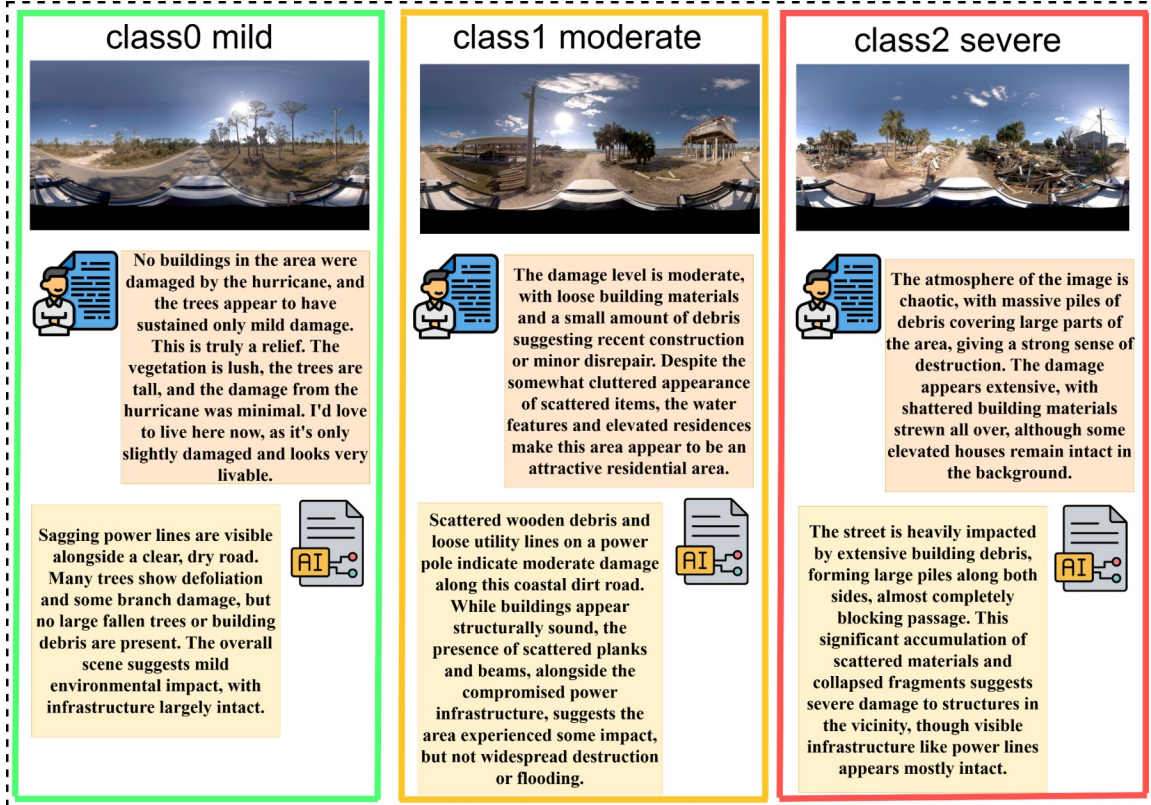


Figure 2. Examples of human- and LLM-generated descriptions of street-view images across different damage severity levels

## 4. Methodology

### 4.1 Experimental Design

The study compared the performance of four architectures for classifying disaster damages using street-view images and fine-tuned pre-trained models (Figure 3). These architectures gradually expand street-view-based disaster damage assessment tasks from unimodal feature extraction to multimodal alignment and disagreement-driven arbitration, aiming to improve overall reliability and performance. Three baseline architectures were designed based on input data modality: image unimodality (Figure 3a), text unimodality (Figure 3b), and image-text multimodality (Figure 3c). We further developed DamageArbiter, a disagreement-driven arbitration framework that leverages the complementary strengths of both unimodal and multimodal models for this task (Figure 3d).



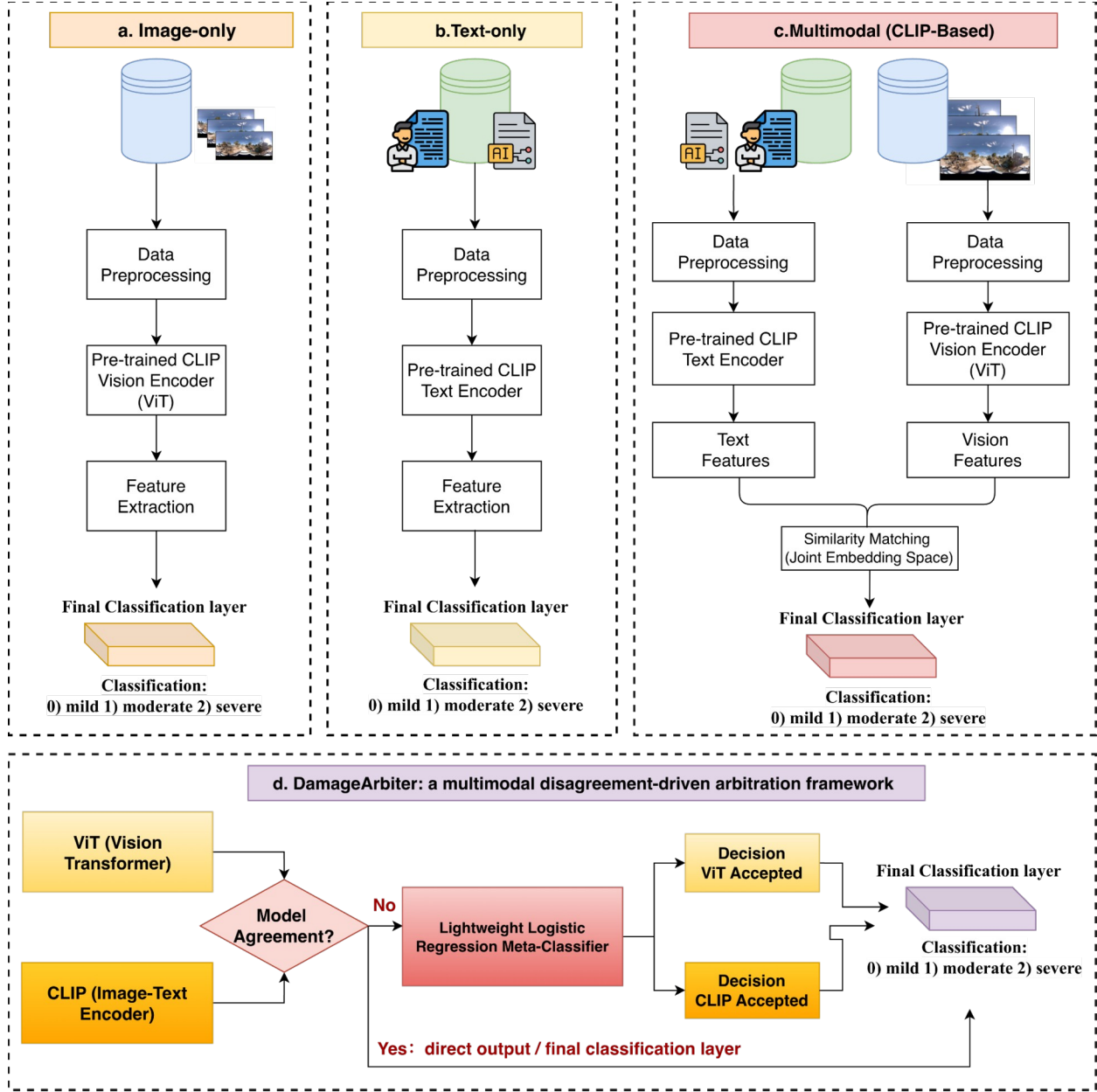


Figure 3. Overall framework of disaster assessment approaches: (a) image-only model, (b) text-only model, (c) multimodal CLIP model, and (d) DamageArbiter, a multimodal disagreement-driven arbitration framework.

Section 4.2 presents the baseline model architectures. [Section 4.3 describes DamageArbiter, including arbitration features, logistic regression training, ablation design, and confidence calibration.](#) Section 4.4 defines the evaluation metrics used to compare models' classification performance, semantic alignment, and reliability. All experiments were implemented in PyTorch using pre-trained models from the timm (<https://huggingface.co/timm>) and CLIP libraries. To ensure a fair comparison between unimodal and multimodal settings, all models were trained and evaluated using consistent data splits and hyperparameter configurations unless otherwise stated.

## 4.2 Baseline Model Architectures and Implementation

### 4.2.1 Unimodal Image-Only Models

We adopted the Vision Transformer (ViT) as the baseline unimodal image-only model for two reasons. First, as a Transformer-based architecture, ViT is designed to model global contextual relationships in images. This capacity is suitable for disaster scenes characterized by spatially distributed and heterogeneous visual patterns. Second, the visual encoder of CLIP is built upon ViT. Using ViT as the unimodal baseline ensures architectural consistency in the experimental design and model comparison.

The ViT model architecture is shown in Figure 4. All street-view images were uniformly scaled to  $224 \times 224$  resolution and normalized before being input into the model. Data augmentation operations, such as random horizontal flipping, were applied during training to improve model generalization. Subsequently, the images were divided into fixed-size patches, which were transformed into sequential inputs via linear projection and position embedding, and then fed into a multi-layer Transformer Encoder. Each encoding layer consisted of multi-head self-attention, a feed-forward network, and layer normalization to capture global dependencies. Finally, the classification token (CLS) output was passed through a multi-layer perceptron (MLP) classification head to achieve a three-category prediction of disaster damage severity.

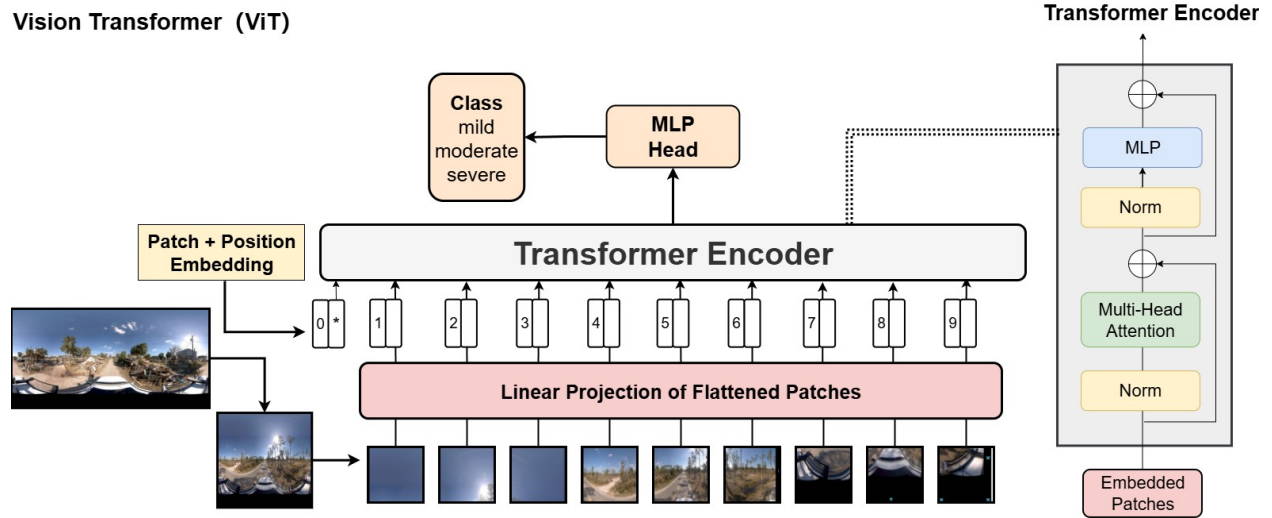


Figure 4. Architecture of the unimodal image-only Vision Transformer (ViT) model for street-view-based disaster damage assessment

We implemented the ViT-Base-Patch16-224 and ViT-Base-Patch32-224 architectures with pre-trained weights from the timm library using PyTorch (Wightman, 2019). The ViT baseline models were fine-tuned using the Milton-SV dataset. The AdamW optimizer was used for parameter updates, with a learning rate of  $1 \times 10^{-4}$ , weight decay of  $1 \times 10^{-4}$ , and a batch size of 32. Cross-entropy loss was used as the objective function, and the maximum number of training epochs was set to 10. To ensure model robustness, we employed a three-fold cross-validation, stratifying the entire dataset according to its class distribution. In each fold iteration, the training set comprised two-thirds of the dataset, and the validation set comprised the other one-third. Out-of-fold (OOF) predictions were also saved for subsequent error and confidence analyses.



### 4.2.2 Unimodal Text-Only Models

The text-only baseline experiment used CLIP’s text encoder as a standalone classifier to isolate and evaluate the contribution of the text modality to disaster damage assessment. We set up two variants. The first, CLIP-Text Encoder (LLM captions), uses disaster descriptions automatically generated by GPT-4o-mini. These descriptions typically have broad, abstract semantic representations that summarize the overall characteristics of the disaster scene. The second, CLIP-Text Encoder (Human captions), uses descriptions written by trained human annotators. These descriptions focus on human-observable, fine-grained disaster signs, such as fallen trees, building debris, and damaged infrastructure. Experiments with LLM-generated input are conducted to test whether synthetic language can serve as an effective input for disaster damage assessment.

In implementation, we used the CLIP text encoder (Transformer architecture, 12-layer encoder, 512-dimensional word embeddings, maximum sequence length of 77) to convert each input text into a 512-dimensional semantic vector, which was then mapped to a three-category space (mild, moderate, and severe) via a linear classifier, as shown in Figure 5. The classification head was trained from scratch on the disaster dataset, while the CLIP text encoder was fine-tuned from pre-trained weights. Both text-only models used the same tokenization and encoding process and were evaluated with three-fold stratified cross-validation under identical hyperparameter settings (learning rate  $1 \times 10^{-4}$ , batch size 32, and AdamW optimizer).

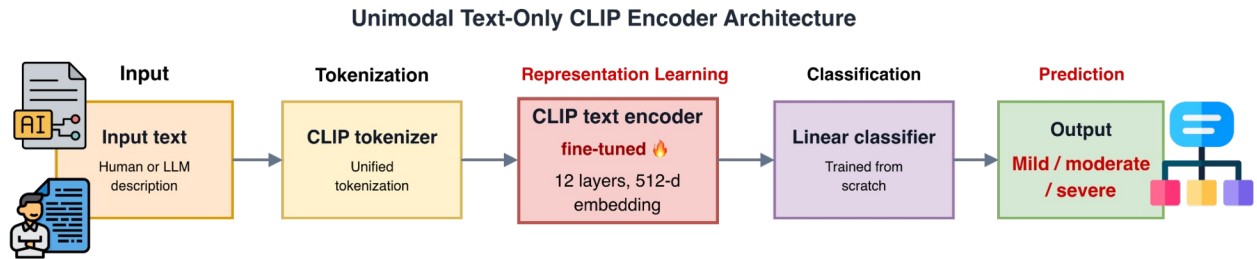


Figure 5. Architecture of the unimodal text-only CLIP encoder for disaster severity classification.

### 4.2.3 Multimodal CLIP Models

To learn robust multimodal representations for disaster assessment, we employed a cross-modal contrastive learning framework based on CLIP. Compared with visual-only or textual-only models, this cross-modal approach can improve interpretability and robustness in complex scenes with class imbalance and ambiguous visual cues. The CLIP-based model architecture is shown in Figure 6. Each fine-tuning sample consists of a post-disaster street-view image and its corresponding text description, either from human annotators or from GPT-4o-mini. This design allows us to compare how human-written and LLM-generated descriptions affect multimodal model learning and disaster assessment.

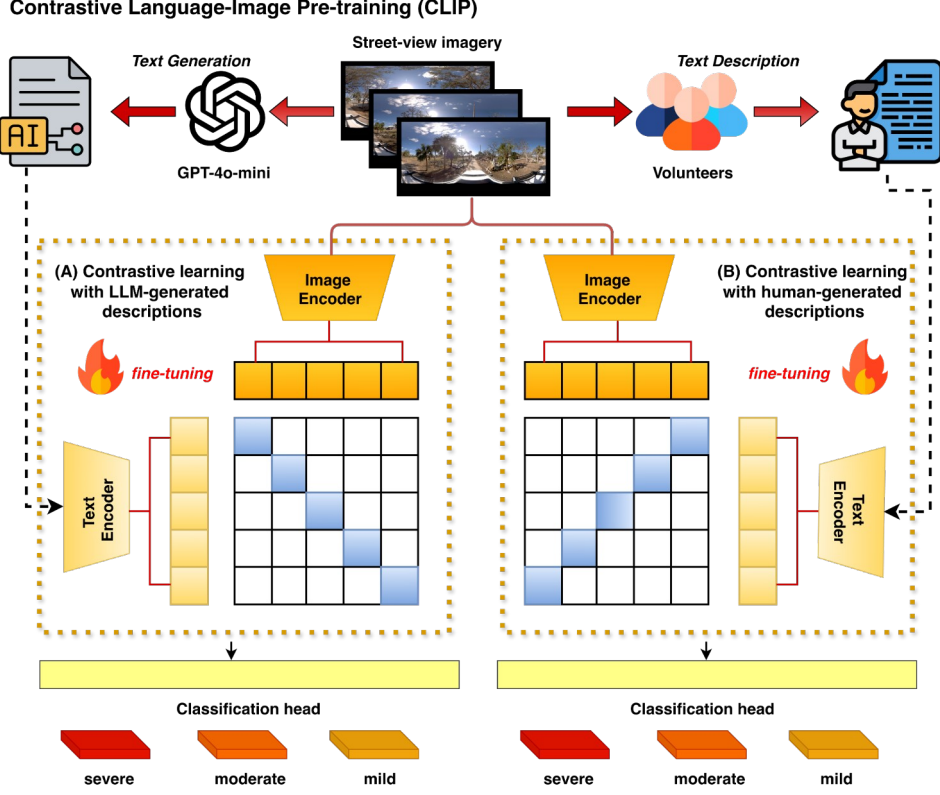


Figure 6. Contrastive Learning for Disaster Assessment with Human vs. LLM-Generated Descriptions.

Visual and textual features are extracted using a pre-trained CLIP model, in which the image is processed by a visual encoder (ViT-B/16) and the text by a transformer-based text encoder (same as the encoder in Section 4.2.2). The resulting embeddings are projected into a joint representation space where cross-modal alignment can be learned through fine-tuning. During training, we optimized a single-directional InfoNCE loss, where each image serves as the query and its associated text description as the positive key. The objective encourages the model to increase similarity between matched image-text pairs while minimizing similarity between mismatched pairs within a batch. This is formulated as:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(x_i, y_i)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(x_i, y_j)/\tau)} \quad (1)$$

where  $x_i$  and  $y_i$  denote the image and text embeddings of the  $i$ -th sample,  $\text{sim}(x, y)$  represents cosine similarity between two normalized embeddings, and  $\tau$  is a temperature scaling factor. Here,  $\tau$  is a learnable temperature inherited from the pre-trained CLIP model (Radford et al., 2021) and fine-tuned jointly with the encoders, rather than a manually set hyperparameter.

To support the downstream classification, we appended a lightweight linear classification head to the combined embedding of the image and text modalities. This head is trained to predict one of three disaster severity levels (mild, moderate, or severe) using the standard cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = -\sum_{i=1}^N \sum_{c=1}^C y_{ic} \log \hat{y}_{ic} \quad (1)$$

where  $C=3$  denotes the number of severity classes, and  $\hat{y}^c_i$  is the predicted softmax probability for class  $c$  of the  $i$ -th sample.

The final training objective combines the contrastive and classification losses as a weighted sum:

$$\mathcal{L}_{\text{total}} = \lambda \mathcal{L}_{\text{contrast}} + (1 - \lambda) \mathcal{L}_{\text{cls}} \quad (1)$$

with a tunable hyperparameter  $\lambda \in [0,1]$  controlling the trade-off between representation alignment and classification accuracy. This hybrid training strategy enables the model to produce representations that are both semantically aligned across modalities and effective for downstream disaster assessment tasks such as damage severity prediction.

This cross-modal contrastive learning framework is designed to improve the robustness and interpretability of disaster assessment models. It also provides a principled setting for evaluating human-written and LLM-generated descriptions, especially when large batches of semantic descriptions are needed for rapid disaster response. This approach lays the foundation for the subsequent analysis of model reliability and error characteristics.

### 4.3 DamageArbiter: A Disagreement-Driven Arbitration Framework

#### 4.3.1 DamageArbiter Model Design

DamageArbiter operates on the outputs of two fine-tuned baseline models, the ViT model described in Section 4.2.1 and the CLIP model described in Section 4.2.3. As shown in Figure 7, DamageArbiter uses a disagreement-driven arbitration strategy rather than directly averaging the ViT and CLIP predictions. For each test instance, the fine-tuned ViT and CLIP models first generate independent damage-category predictions. When the two models predict the same category, DamageArbiter uses this shared output as the final prediction. When the two base models disagree, a logistic regression arbitrator is activated to determine which model prediction is more reliable and should be adopted. [As a final step, the confidence of the adopted prediction is recalibrated through post-hoc temperature scaling \(Section 4.3.4\); because this rescales confidence without changing the predicted class, it mitigates overconfidence while leaving the arbitration decision and the accuracy unchanged.](#)

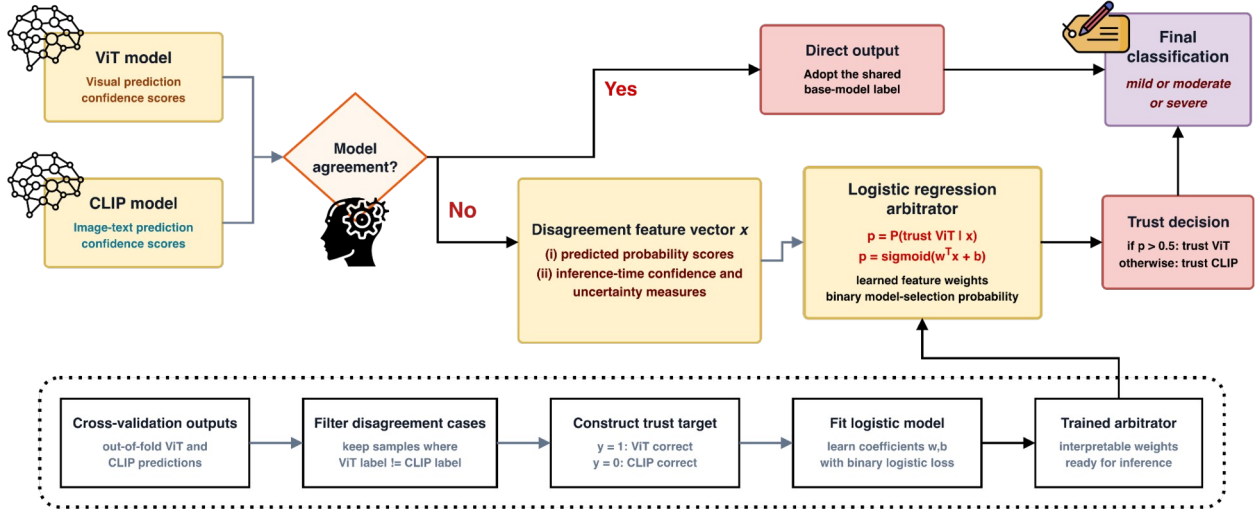


Figure 7. Logistic regression-based DamageArbiter for disagreement-driven ViT-CLIP arbitration.

For each sample, the arbitration feature vector uses inference-time confidence and uncertainty measures derived from each model’s predicted probability distribution over the three damage categories (mild, moderate, and severe): the max-softmax confidence (the probability of each model’s own predicted class), Shannon entropy, and decision margin (top-1 minus top-2 probability), all computed at inference time (Section 4.3.2). These measures quantify how confident or uncertain a prediction is and whether a disagreement arises from misplaced confidence or from genuine class ambiguity. We conducted ablation experiments (Section 4.3.3) to determine the final feature set used by the logistic regression arbitrator.

#### 4.3.2 Predictive Confidence and Uncertainty Measures

We characterized each model’s prediction using four measures derived from its predicted probability distribution over the three damage categories: two confidence-based error types (overconfidence and ambiguity) and two continuous uncertainty metrics (Shannon entropy and decision margin). Together, these measures describe not only how uncertain a prediction is, but also whether an incorrect prediction stems from misplaced confidence or from poor class separability.

The two error types characterize the confidence level of incorrect predictions and were estimated through a confidence-based analysis of misclassified samples (Figure 8). For each misclassified sample, we computed the confidence margin as the difference between the model’s predicted probability for the incorrectly predicted class and its predicted probability for the ground truth class. A high confidence margin (0.4~1.0) indicates that the model was strongly confident in an incorrect prediction; such samples were labeled overconfidence errors. A low confidence margin (0~0.1) indicates that the predicted probabilities of the incorrect and true classes were nearly equal; such samples were labeled ambiguity errors. The remaining misclassified samples (confidence margin 0.1~0.4) were categorized as medium-confidence errors. From a disaster-assessment perspective, overconfidence errors are particularly problematic because they reflect false certainty in incorrect predictions, whereas ambiguity errors are informative because they reveal cases of low-class separability, which may correspond to visually ambiguous damage patterns or insufficient evidence.

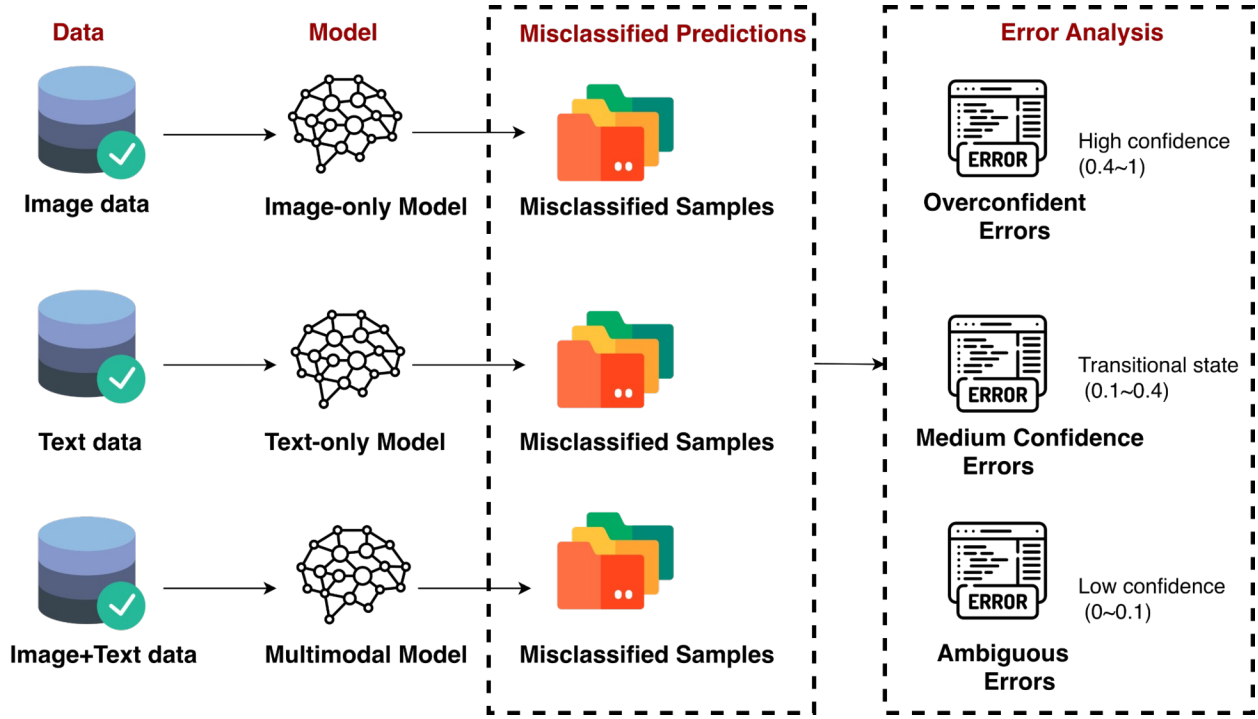


Figure 8. Framework of confidence-based error analysis in disaster assessment models.

The two continuous uncertainty metrics, entropy and decision margin, are computed directly from the full predicted probability distribution and therefore apply to every prediction, not only to misclassified samples.

Entropy quantifies how spread out the probability mass is across categories and is calculated as follows. Let  $p = (p_1, p_2, \dots, p_C)$  denote a model's predicted probability distribution over the  $C$  damage categories (here  $C = 3$ ), where  $p_i$  is the predicted probability of category  $i$  and  $\sum p_i = 1$ . The Shannon entropy of this distribution is:

$$H(p) = - \sum_i p_i \log p_i$$

Entropy reaches its minimum ( $H = 0$ ) when the model assigns all probability to a single category (a fully confident, one-hot prediction), and its maximum ( $H = \log C$ ) when the predicted probabilities are uniformly distributed across the  $C$  categories (33% probability for each category in this case study). Higher entropy, therefore, indicates greater predictive uncertainty. We used the natural logarithm in this experiment, and any fixed logarithm base only rescales  $H$  by a constant factor.

The decision margin is calculated using Equation X. Let  $p_{(1)}$  and  $p_{(2)}$  denote the highest and second-highest predicted probabilities for a given sample (i.e., the top-1 and top-2 class probabilities). The decision margin is their difference:

$$m = p_{(1)} - p_{(2)}$$

A small margin means the two most likely categories are nearly tied, indicating weak class separability and high prediction ambiguity; a large margin means the top category is clearly separated from the rest, indicating a confident, well-separated decision.

### 4.3.3 Logistic Regression Training and Feature Selection

Logistic regression was selected because it provides a lightweight and reproducible arbitration model suitable for the relatively small set of disagreement samples. The logistic regression arbitrator was trained using disagreement cases derived from the cross-validation outputs of the ViT and CLIP baseline models. For each fold, predictions were generated by models not trained on the corresponding validation samples, ensuring that the arbitrator was trained on out-of-fold predictions. We then identified validation samples for which ViT and CLIP produced different damage level predictions. The arbitration target was defined as a binary label. Samples were labeled as ( $y = 1$ ) when the ViT prediction matched the ground truth and labeled as ( $y = 0$ ) when the CLIP prediction matched the ground truth. Disagreement samples for which neither base model predicts the correct label were excluded from arbitrator training.

Given a disagreement feature vector  $x$ , the logistic regression model estimates the probability that the final decision should trust ViT rather than CLIP as:

$$P(y = 1 \mid x) = \sigma(w^T x + b),$$

where  $\sigma$  denotes the sigmoid function,  $w$  is the coefficient vector, and  $b$  is the intercept. At inference time, let  $g(x) = P(y = 1 \mid x)$  be the probability produced by the logistic regression arbitrator, where  $x$  is the arbitration feature vector and  $y = 1$  denotes that the ViT prediction should be trusted. When ViT and CLIP disagree, the final prediction follows ViT if  $g(x) > \tau$  and follows CLIP otherwise, where  $\tau$  is the arbitrator’s decision threshold.

To identify the final feature set for the arbitrator, we conducted ablation experiments using progressively richer feature configurations. The configurations started with each model’s predicted probability for its selected class, or max-softmax confidence (LOGREG\_conf), and then incrementally added uncertainty measures, including entropy and decision margin (LOGREG\_conf+unc). We also included a non-learned heuristic that selects the prediction from the more confident model as a deployable reference.

All feature configurations were evaluated using the same selected disagreement samples and binary trust-label definition described above. For each configuration, a logistic regression arbitrator was trained using only features that can be computed without ground-truth labels at inference, including predicted-class confidence, entropy, and decision margin. The decision threshold was fixed at ( $\tau = 0.5$ ) across all configurations to avoid per-configuration threshold tuning, isolate the contribution of each feature group, and enable a fair comparison. After training, the logistic regression coefficients and decision threshold can be frozen and applied to newly collected, unlabeled street-view imagery using the same feature-computation procedure. These ablation experiments quantify the contribution of confidence and uncertainty features to disagreement-driven arbitration and provide the empirical basis for selecting the optimal DamageArbiter configuration.



#### 4.3.4 Confidence Calibration

Because overconfidence reflects a model assigning probabilities that exceed its empirical accuracy (Guo et al., 2017), we calibrated the image model using post-hoc temperature scaling. Given the logits  $z = (z_1, \dots, z_C)$  that the model produces for the  $C$  damage categories, temperature scaling divides them by a single positive scalar  $T$  before the softmax:

$$p_i = \exp(z_i / T) / \sum_j \exp(z_j / T)$$

The temperature  $T$  was fit on out-of-fold validation predictions by minimizing the negative log-likelihood and was then applied at inference time. Dividing all logits by the same  $T$  preserves their order, so the predicted class, and therefore the classification accuracy, is unchanged; only the confidence values are rescaled, with  $T$  greater than one reducing overconfidence and  $T$  less than one sharpening it.

We assessed calibration quality using the expected calibration error (ECE). Predictions are grouped into  $M$  equal-width confidence bins, and the ECE is the average gap between accuracy and confidence across bins, weighted by the number of predictions in each bin:

$$\text{ECE} = \sum_m (|B_m| / N) \cdot |\text{acc}(B_m) - \text{conf}(B_m)|$$

where  $B_m$  is the set of predictions whose confidence falls in bin  $m$ ,  $|B_m|$  its size,  $N$  the total number of predictions, and  $\text{acc}(B_m)$  and  $\text{conf}(B_m)$  the mean accuracy and mean confidence within the bin. We also report the proportion of overconfident errors (defined in Section 4.3.2) before and after calibration. Calibration is applied as a post-hoc step to the base-model probability outputs and does not modify the arbitration decisions defined in Section 4.3.1, so it leaves the predicted classes and the overall accuracy unchanged. The effect of calibration on the ViT model and on DamageArbiter is reported in Section 5.3.

#### 4.4 Evaluation Metrics

We evaluated the tested models in two steps. First, the baseline models were evaluated and compared using standard classification-based metrics and further analyzed using reliability measures to characterize confidence-based error patterns. In this step, image-text semantic alignment was also evaluated to assess the consistency between human- and LLM-generated annotations and the corresponding post-disaster street-view images. Second, the optimal DamageArbiter model configuration, identified through ablation experiments, was compared with the baseline models in terms of both classification accuracy and reliability.

Classification performance was evaluated using Accuracy, Recall, Precision, sample-weighted F1-score (SW-F1), and Matthews Correlation Coefficient (MCC), which were computed based on the confusion matrix components: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). The classification metrics were computed using equations [X-X](#). These metrics were used to compare the baseline models and the final selected DamageArbiter model. SW-F1 was included to account for class imbalance across damage categories, whereas MCC was used as a balanced metric that considers all components of the confusion matrix.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (10)$$

$$\text{SW-F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (12)$$

Semantic alignment was evaluated using CLIPScore, a reference-free image-text alignment metric derived from CLIP. For each image-text pair, CLIP encodes the image (I) and text description (T) into embeddings  $\mathbf{v}_I \in \mathbb{R}^d$  and  $\mathbf{v}_T \in \mathbb{R}^d$ , where  $d$  is the dimensionality of the shared multimodal embedding space. CLIPScore was computed as the cosine similarity between the normalized image and text embeddings (Equation X), with higher values indicating stronger semantic consistency. In this study, CLIPScore was used to compare the semantic alignment between human or LLM-generated annotations and the corresponding street-view images. A higher CLIPScore indicates stronger semantic consistency between the image and its description.

$$\text{CLIPScore}(I, T) = \cos(\mathbf{v}_I, \mathbf{v}_T) = \frac{\mathbf{v}_I \cdot \mathbf{v}_T}{\|\mathbf{v}_I\| \|\mathbf{v}_T\|} \quad (1)$$

Model uncertainty was evaluated using the confidence-based error categories defined in Section 4.3.2, including overconfident, medium-confidence, and ambiguous errors. Class-specific misclassification rates were also reported for mild, moderate, and severe damage categories to examine whether different baseline models showed systematic error patterns across damage levels.

DamageArbiter feature selection was evaluated through ablation experiments on disagreement samples. Each logistic regression arbitrator configuration was evaluated using Accuracy and MCC on the full test set. After the final feature configuration was selected, the resulting DamageArbiter model was evaluated against the baseline models using the same classification metrics: Accuracy, Recall, Precision, SW-F1, and MCC.

## 5. Results

### 5.1 Baseline Model Performance and Reliability

Table 1 summarizes the performance of all baseline models across five classification metrics, along with the CLIPScore values for the two baseline CLIP models. Overall, the image-only ViT-B/32 model achieved

the strongest baseline performance across most metrics, including accuracy (0.7433), recall (0.7433), precision (0.7597), and MCC (0.5947). CLIP with human annotations achieved the highest SW-F1 score (0.7403), although its advantage over ViT-B/32 was marginal. The text-only baselines performed the worst across all five metrics, indicating that converting post-disaster images to text may omit visual cues that are important for distinguishing damage levels.

Table 1. Comparative Evaluation of image-only, text-only, and multimodal models on Disaster Severity Classifications

| Model                | Modality   | Accuracy      | Recall        | Precision     | SW-F1         | MCC           | Overconfident | Medium        | Ambiguous     | CLIPScore     |
|----------------------|------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| ViT-B/16             | Image Only | 0.7261        | 0.7261        | 0.7320        | 0.7240        | 0.5653        | 64.86%        | 27.14%        | 8.00%         | -             |
| ViT-B/32             | Image Only | <b>0.7433</b> | <b>0.7433</b> | <b>0.7597</b> | 0.7386        | <b>0.5947</b> | <b>70.58%</b> | 21.04%        | 8.38%         | -             |
| Text Encoder (LLM)   | Text Only  | 0.6307        | 0.6307        | 0.6331        | 0.6261        | 0.4126        | 47.78%        | 37.71%        | 14.51%        | -             |
| Text Encoder (Human) | Text Only  | 0.6710        | 0.6710        | 0.6779        | 0.6715        | 0.4788        | 53.27%        | 33.65%        | 13.08%        | -             |
| CLIP (LLM)           | Image+Text | 0.7238        | 0.7238        | 0.7249        | 0.7229        | 0.5644        | 0.00%         | 55.18%        | <b>44.82%</b> | 0.2467        |
| CLIP (Human)         | Image+Text | 0.7422        | 0.7422        | 0.7504        | <b>0.7403</b> | 0.5915        | 0.00%         | <b>66.82%</b> | 33.18%        | <b>0.2701</b> |

*Note: Bold values indicate the best-performing model for each classification metric.*

For the multimodal CLIP baselines, CLIP-Human outperformed CLIP-LLM across all classification metrics, including accuracy (0.7422 vs. 0.7238), SW-F1 (0.7403 vs. 0.7229), and MCC (0.5915 vs. 0.5644). CLIP-Human also achieved a higher CLIPScore than CLIP-LLM (0.2701 vs. 0.2467), indicating stronger semantic alignment between human annotations and the corresponding street-view images. Consistently, the text-only CLIP model with human-written descriptions outperformed the version with LLM-generated descriptions. However, CLIP-Human remained slightly below ViT-B/32 in accuracy and MCC, suggesting that the strongest baseline performance depended primarily on visual evidence.

The confidence-based error profiles in Table 1 reveal a different pattern. The image-only ViT models produced the highest proportions of overconfident errors, with 64.86% of ViT-B/16 errors and 70.58% of ViT-B/32 errors falling into the high-confidence error range. This indicates that the best-performing image-only baseline also produced the largest number of highly confident incorrect predictions. In contrast, the multimodal CLIP models produced no overconfident errors. Their errors were concentrated in the medium-

confidence and ambiguous ranges, particularly for CLIP-LLM, for which ambiguous errors accounted for 44.82% of all errors.

Overall, Table 1 shows that accuracy and reliability do not necessarily improve in parallel. Although ViT-B/32 achieved the best classification accuracy among the baseline models, it also produced the highest proportion of overconfident errors. In contrast, the multimodal CLIP models reduced overconfidence but still exhibited ambiguity and class-specific misclassification. These findings motivate the DamageArbiter framework, which uses disagreement-based arbitration to integrate the strong visual classification performance of ViT with the more conservative reliability profile of CLIP.

## 5.2 Ablation Study and Performance Evaluation of DamageArbiter

Table 3 summarizes the ablation: we ablate the arbitrator’s feature set using only label-free, inference-time features, at a fixed decision threshold  $\tau = 0.5$ . For each configuration we report the accuracy on the full test set and MCC. All learned configurations perform comparably (accuracy 0.7580–0.7585; MCC 0.618–0.619) and clearly exceed the non-learned heuristic that simply trusts the more confident model (0.7504; MCC 0.6063), confirming that the logistic regression extracts signal a fixed rule cannot. The differences among learned configurations are within run-to-run variation: adding uncertainty features does not yield a meaningful improvement over the max-softmax confidence alone. We therefore select the confidence-only configuration as the final arbitrator (accuracy 0.7585; MCC 0.6188) for parsimony and deployability.

**Table 2. Ablation study of logistic regression arbitration with different feature configurations.**

| Arbitration Setting    | Accuracy | MCC    |
|------------------------|----------|--------|
| <b>LOGREG_conf</b>     | 0.7585   | 0.6188 |
| <b>LOGREG_conf+unc</b> | 0.7580   | 0.6181 |
| <b>Naive heuristic</b> | 0.7504   | 0.6063 |

The ablation also clarifies when each base model should be trusted. When the confidence features indicate that one model is markedly more confident and its predicted distribution is stable (low entropy, large decision margin), that model is prioritized: ViT is favored for its strong visual discrimination, whereas CLIP is favored when ViT shows signs of overconfidence or when the disagreement arises under high uncertainty, because CLIP’s cross-modal alignment reduces misleading visual predictions. This behavior is learned by the logistic-regression arbitrator from the out-of-fold disagreement samples rather than imposed as a fixed rule, and it aligns with the reliability requirements of disaster assessment: ViT provides high-resolution visual discrimination, while CLIP provides robustness against ambiguous visual cues, and their arbitrated combination yields more reliable decisions on divergent samples.

Figure 9 compares the optimal DamageArbiter with the strongest baseline on each evaluation metric. DamageArbiter outperforms the best baseline across all metrics, although the gains on the accuracy-type metrics are modest: accuracy improves from 0.7433 to 0.7585, recall from 0.7433 to 0.7585, precision from 0.7597 to 0.7713, and SW-F1 from 0.7403 to 0.7550 (absolute gains of 0.0152, 0.0152, 0.0116, and 0.0147, respectively). These improvements are consistent across the full metric set rather than driven by any single metric.

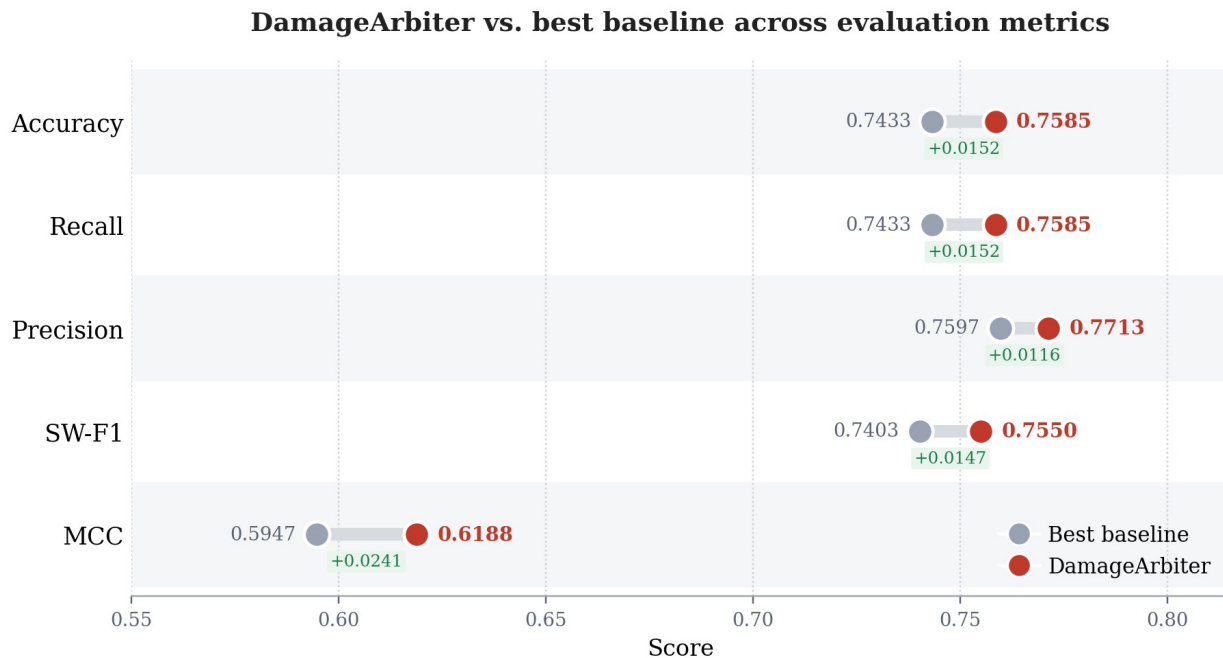


Figure 9. DamageArbiter versus the best baseline across five evaluation metrics (per-metric improvement).

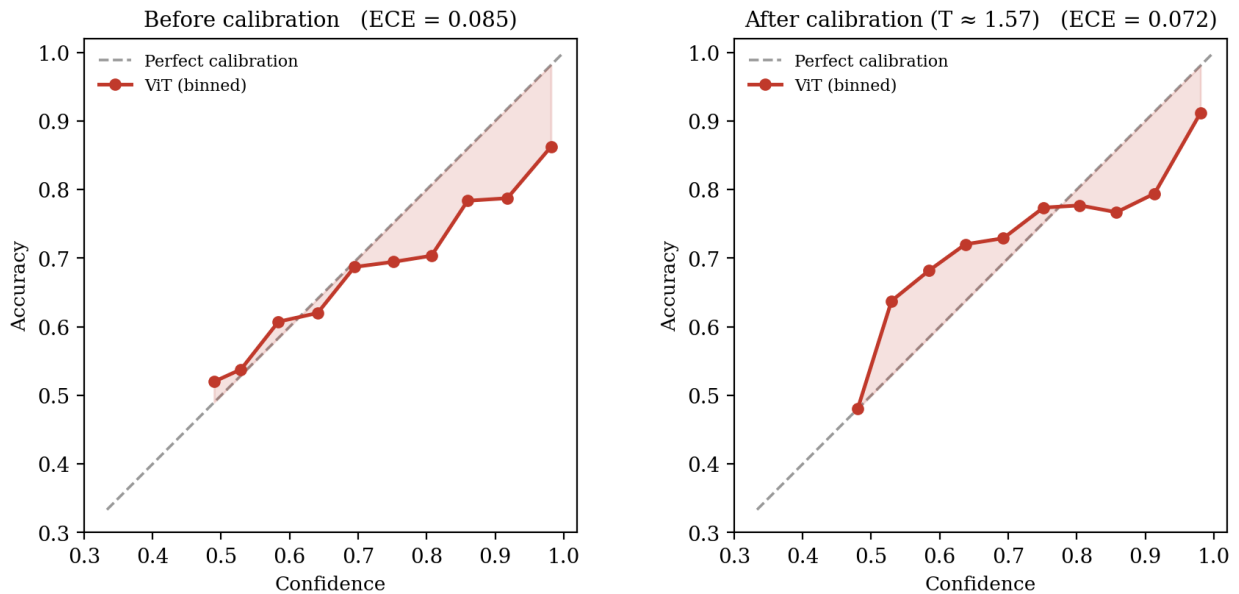
The largest improvement appears on the Matthews correlation coefficient (MCC), which increases from 0.5947 to 0.6188, an absolute gain of 0.0241. Because MCC accounts for all entries of the confusion matrix and is sensitive to class imbalance, this gain indicates that DamageArbiter yields more balanced and reliable predictions across damage categories rather than improving only the majority class. As Figure 9 shows, the MCC gap between DamageArbiter and the best baseline is proportionally larger than the gains on the other metrics, suggesting that the arbitration mechanism is particularly effective at correcting errors that most severely degrade balanced classification quality.

Following the same confidence-based criterion used for the baseline models (Section 4.3.2), we also quantified overconfidence for DamageArbiter by attributing each final prediction to the model the arbitrator adopted (ViT when the arbitrator trusts ViT, CLIP otherwise; agreement cases inherit the shared prediction). Under this definition, 74.4% of DamageArbiter’s remaining errors are overconfident, comparable to the ViT-B/32 baseline (70.58%) rather than to CLIP (0%). This pattern is expected: because the confidence-only arbitrator follows ViT on 93% of samples (2,389 of 2,556), it largely inherits ViT’s confidence profile, and most of its residual errors are cases in which ViT and CLIP agreed on the same

wrong class. Accordingly, the contribution of DamageArbiter is an improvement in accuracy and balanced classification (MCC) together with an explicit diagnosis of overconfidence, rather than a reduction of overconfidence itself. We address overconfidence directly in Section 5.3 through post-hoc confidence calibration, which reduces it without sacrificing the accuracy gains reported above.

### 5.3 Confidence Calibration

Section 5.2 showed that DamageArbiter inherits the overconfidence of its image backbone. We therefore applied to the image model the post-hoc temperature scaling described in Section 4.3.4. The fitted temperature was  $T \approx 1.57$  (greater than one, confirming that ViT was systematically overconfident). As shown in Figure 10, calibration brings the high-confidence predictions closer to the diagonal of perfect calibration and lowers the expected calibration error (ECE) from 0.085 to 0.072. More importantly for disaster assessment, calibration reduces the proportion of overconfident errors from 70.58% to 51.37% for ViT and, correspondingly, from 74.4% to 54.7% for DamageArbiter, which adopts the calibrated ViT outputs. Because temperature scaling preserves the predicted class, calibration is applied as a post-hoc step on the model outputs and leaves the arbitration decisions, and therefore the accuracy (75.85%) and MCC (0.619), unchanged. Calibration thus composes with the arbitration framework: arbitration improves accuracy, calibration improves reliability, and the two can be applied together. The residual overconfident errors (54.7%) correspond largely to cases in which ViT and CLIP agree on the same incorrect class, which neither arbitration nor calibration can resolve.





*Figure 10. Reliability diagrams of the ViT model before and after temperature scaling ( $T \approx 1.57$ ). Calibration lowers the expected calibration error (ECE) and brings high-confidence predictions closer to perfect calibration.*

#### **5.4 Spatial Visualization**

We deployed the selected DamageArbiter to every street-view location to examine the spatial structure of the predicted damage, the arbitration behavior, and the model's errors (Figure 11). The predicted-severity map (Figure 11b) closely reproduces the ground-truth pattern (Figure 11a), with the most severe damage concentrated along the exposed coastal and central streets. The arbitration-decision map (Figure 11c) shows where the disagreement-driven mechanism was engaged: the two base models agreed at the majority of locations ( $n = 2,041$ ), whereas disagreements clustered along the main interior corridors, where the arbitrator retained the ViT prediction in 348 cases and adopted the CLIP prediction in 167. Finally, the error map (Figure 11d) shows that residual misclassifications are spread across the network rather than confined to one area, and that overconfident errors, rather than ambiguous ones, dominate, consistent with the confidence analysis in Section 5.2.

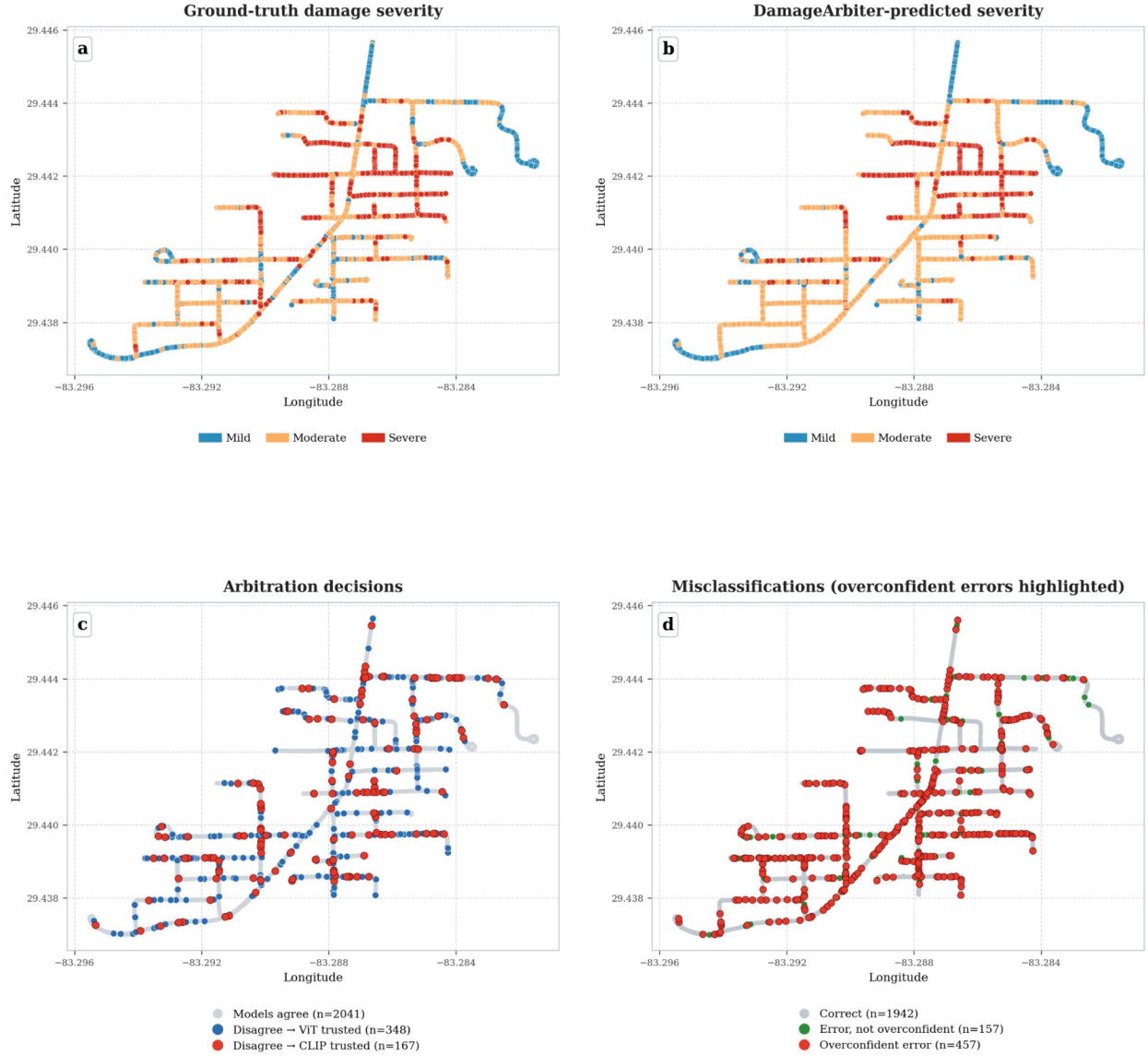


Figure 11. Spatial deployment of DamageArbiter across street-view locations in Horseshoe Beach, Florida. (a) Ground-truth damage severity; (b) DamageArbiter-predicted severity; (c) arbitration decisions, distinguishing locations where ViT and CLIP agreed from disagreement cases resolved in favor of ViT or of CLIP; (d) misclassified locations, with overconfident errors highlighted.

## 6. Discussions

This study proposes and evaluates DamageArbiter, a multimodal post-disaster assessment framework based on disagreement-driven arbitration. The results show that image-based models such as ViT provide strong visual discrimination for damage-severity classification, but their errors can be highly confident, whereas CLIP-based multimodal models show a more conservative reliability profile and provide a semantic interface for interpreting disaster-related visual cues. By using arbitration only when ViT and CLIP disagree, DamageArbiter combines the discriminative strength of image-based classification with the semantic interpretability of language-image modeling, improving overall accuracy and balanced

classification (MCC). However, because the selected confidence-only arbitrator follows the image model on the large majority of samples, it inherits the image model’s confidence profile and does not by itself reduce overconfident errors; its reliability contribution is therefore diagnostic, quantifying where and how overconfidence arises, rather than corrective. We show in Section 5.3 that this overconfidence is instead corrected by post-hoc calibration, which composes with the arbitration framework without affecting its accuracy. Compared with the best baseline for each metric, DamageArbiter improved accuracy from 0.7433 to 0.7585, recall from 0.7433 to 0.7585, precision from 0.7597 to 0.7713, SW-F1 from 0.7403 to 0.7550, and MCC from 0.5947 to 0.6188, suggesting that disagreement-driven arbitration improves predictive performance and balanced classification, while the accompanying reliability analysis quantifies, rather than removes, the overconfidence behavior of each model.

The contributions of this work are threefold. First, it shows that accuracy alone is insufficient for evaluating classification-based damage-assessment models: an image-only model can attain competitive accuracy while committing a high proportion of overconfident errors (over 70%), so overconfidence should be quantified and reported alongside accuracy. Second, it proposes DamageArbiter, a disagreement-driven framework that combines unimodal and multimodal models, improving accuracy and balanced classification (MCC) over the strongest baselines while making each model’s overconfidence behavior explicit and interpretable through confidence analysis. Third, it shows that the resulting overconfidence can be substantially reduced through post-hoc calibration without sacrificing accuracy, identifying calibration, rather than arbitration, as the appropriate mechanism for improving reliability.

This study also has two main limitations. First, the experiments focus on a single hurricane event and a limited geographic area, so the generalizability of the results to other regions, disaster types, and urban forms remains to be tested. Second, DamageArbiter currently assesses damage by severity level (mild, moderate, and severe) rather than by damage type, and does not yet produce type-specific damage maps; we identify type-specific mapping as a direction for future work.

These contributions and limitations point to three directions for future work. First, to move beyond the single-event, single-region scope, the framework should be evaluated and adapted across additional hurricane events, geographic regions, disaster types, and urban forms; where labeled post-disaster imagery is scarce, data augmentation, including synthetic post-disaster street-view generation, together with domain-adaptation techniques, could improve transferability. Second, to advance beyond severity-only assessment, the framework can be extended from severity classification toward damage-type mapping, distinguishing fallen trees, building debris, infrastructure disruption, and flooding, using open-vocabulary or zero-shot detection and segmentation with vision-language models, so that it produces the type-specific damage maps that operational response requires. Third, building on our finding that overconfidence is best addressed through calibration, reliability can be integrated more deeply into the framework: future work could pursue training-time calibration, such as label smoothing or focal loss, and uncertainty-aware arbitration, and could exploit the calibrated confidence for human-in-the-loop deployment, automatically flagging low-confidence assessments for field verification while accepting high-confidence predictions. Together, these directions would extend DamageArbiter from a reliable severity-classification framework toward a transferable, type-aware, and operationally trustworthy disaster-assessment workflow.

## 7. Conclusion

This study develops DamageArbiter, a disagreement-driven multimodal arbitration framework for street-view-based post-disaster damage assessment. We systematically compare unimodal image, unimodal text, and multimodal (CLIP) models against the proposed arbitration model for disaster severity classification. The image model (ViT) exhibits strong discriminative power but is prone to overconfidence errors; the text model adds complementary semantic information yet remains unstable; and CLIP improves robustness, particularly in high-uncertainty scenes. **By arbitrating between the complementary strengths of ViT and CLIP, DamageArbiter raises overall accuracy to 75.85% and MCC to 0.619, the highest among all deployable models, using only features computable without ground-truth labels, while a complementary reliability analysis quantifies the overconfidence behavior of each model. Post-hoc calibration further reduces overconfident errors from over 70% to roughly 54% without sacrificing accuracy, yielding a model that is both more accurate and better calibrated.**

Beyond accuracy, our confidence analysis exposes distinct error patterns rooted in overconfidence and class ambiguity, offering actionable guidance for model selection and human-machine collaboration in emergency decision-making. Taken together, these results demonstrate that disagreement-driven arbitration is an effective and practical approach for street-view-based disaster assessment, and provide methodological support for building more reliable and interpretable disaster-intelligence systems.

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## Disclosure statement

No potential conflict of interest was reported by the authors.

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